

Srikar Katta

PHD CANDIDATE · APPLE SCHOLAR IN AI/ML · CAUSAL INFERENCE · INTERPRETABLE MACHINE LEARNING

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Summary

I develop methods for reliable inference and interpretation in complex data environments across the health and social sciences. To this aim, I leverage generative AI and modern machine learning to estimate causal effects when we have complex data from social media environments [7] or wearable devices [3]. Because the reasoning-process behind many of these generative AI and machine learning systems are opaque, I also develop benchmarking and explainability tools to audit the capabilities and limitations of these black boxes [1, 8]. My work has been recognized by multiple student paper awards from the American Statistical Association, spotlights from leading machine learning conferences, and the Apple Scholar in AI/ML Fellowship.

Education

Duke University

PHD IN COMPUTER SCIENCE

- Advisors: Cynthia Rudin & Alexander Volfovsky

Durham, NC

August 2022 - PRESENT

Temple University

BS IN MATHEMATICS & COMPUTER SCIENCE

- Minor in Economics

Philadelphia, PA

August 2018 - May 2022

GPA: 3.96/4.00

Honors & Awards

- 2026 **Risk Analysis Section Early Career Paper Award Honorable Mention**, American Statistical Association
 - Awarded for quantifying variable importance even in the presence of unobserved variables
- 2025 **Apple Scholar in AI/ML Fellowship**, Apple
 - Invitation-only, selective fellowship supporting emerging leaders in machine learning research
- 2024 **Biometrics Section Early Career Paper Award**, American Statistical Association
 - Awarded for designing interpretable estimators of causal effects from wearables and sensor data
- 2023 **1st Place**, AAAI W3PHIAI Hackathon on Aging
 - Awarded for highlighting the distribution shift issues arising from biological age predictors
- 2023 **Open Data Award**, Scientific Research Network on Decision Neuroscience and Aging
 - Awarded for creating high-quality, publicly available data for studying financial exploitation
- 2022 **Scott Hibbs Future of Computing Award**, Temple University
 - Awarded to a computer science student at Temple University demonstrating academic excellence

Publications

PUBLISHED WORK

* indicates joint authorship

- [1] J. Donnelly*, **S. Katta***, E. Borgonovo, and C. Rudin, "Variable importance with unobserved confounding and the rashomon effect," *International Conference on Artificial Intelligence and Statistics*, 2026, **Risk Analysis Section Early Career Paper Award Honorable Mention, American Statistical Association**.
- [2] H. Jeong*, **S. Katta***, W. K. Wang, A. Volfovsky, and J. Dunn, "Impact of daylight saving time on physical activity patterns," *Nature Health*, 2026.
- [3] **S. Katta**, H. Parikh, C. Rudin, and A. Volfovsky, "Interpretable causal inference for analyzing wearable, sensor, and distributional data," *International Conference on Artificial Intelligence and Statistics*, pp. 3340–3348, 2024, **Biometrics Section Early Career Paper Award, American Statistical Association**.

- [4] C. Rudin, C. Zhong*, L. Semenova*, M. Seltzer*, R. Parr*, J. Liu*, **S. Katta***, J. Donnelly*, H. Chen*, and Z. Boner*, “Amazing things come from having many good models,” *International Conference of Machine Learning*, 2024, **Spotlight Paper (Top 3% of Submissions)**.
- [5] J. Donnelly*, **S. Katta***, C. Rudin, and E. Browne, “The Rashomon importance distribution: Getting rid of unstable, single model-based variable importance,” *Advances in Neural Information Processing Systems*, vol. 36, pp. 6267–6279, 2023, **Spotlight Paper (Top 3% of Submissions)**.
- [6] Y. J. Yang, K. Hackett, **S. Katta**, R. M. Ludwig, J. Jarcho, T. Giovannetti, D. S. Fareri, and D. V. Smith, “Psychological, social, and health-related factors predict risk for financial exploitation,” *Communications Psychology*, 2023, **SRNDNA Open Data Award**.

SUBMITTED WORK

- [7] A. van Loon, **S. Katta**, C. A. Bail, D. S. Hillygus, and A. Volfovsky, “Designing social media to promote productive political dialogue on a new research platform,” Under Review at Nature Scientific Reports, 2025.
- [8] G. Tierney*, **S. Katta***, C. Bail, S. Hillygus, and A. Volfovsky, “A new experimental design for causal inference with text: Can a language model be too big?” Revise and resubmit at JRSSA, 2025.

IN PREPARATION

- [9] **S. Katta**, H. Parikh, C. Rudin, A. Volfovsky, H. Sun, and B. Westover, *Why machine learning your biological age fails*, **1st place prize in AAAI W3PHIAI Hackathon on Aging**, 2025+.

Technical Experience

Apple, Inc

New York City, NY

MACHINE LEARNING RESEARCH SCIENTIST INTERN

May 2025 – Aug. 2025

- Developed longitudinal biomarkers of aging from over 149 million participant-days of wearables data
- Improved disease diagnoses of cross-sectional aging biomarkers by 20%+
- Predicted onset of adverse heart events with same accuracy as invasive longitudinal biomarkers

Amazon

Seattle, WA

APPLIED SCIENTIST INTERN

May 2024 – Aug. 2024

- Automated analyses of customer-level experimental data at scale, reducing runtime by at least 10x.
- Analyzed *hundreds* of customer-level datasets with over 9 billion observations to generate causal insights under unobserved confounding.
- Derived asymptotic behavior of new estimator, enabling efficient computation of p-values and confidence intervals.

Invited Presentations

INVITED TALKS

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| 2025 | Duke University , Guest Lecture for Causal Inference in Data Analysis with Applications to Fairness and Explanations | <i>Durham, NC</i> |
| 2025 | INRIA , Interpretable causal inference for analyzing wearable, sensor, and distributional data | <i>Paris, FR</i> |
| 2025 | Statistical Learning and Data Science , Interpretable causal inference for analyzing wearable, sensor, and distributional data | <i>Newport Beach, CA</i> |
| 2024 | Joint Statistics Meeting , Interpretable causal inference for analyzing wearable, sensor, and distributional data | <i>Portland, OR</i> |
| 2024 | Blue Cross Blue Shield of North Carolina , Interpretable causal inference via almost matching exactly | <i>Virtual</i> |
| 2024 | INFORMS , Interpretable causal inference for analyzing wearable, sensor, and distributional data | <i>Phoenix, AZ</i> |

POSTER PRESENTATIONS

- 2025 **Experimental Design in the Age of Generative AI**, A new experimental design for causal inference with text: Can a language model be too big? *Berkeley, CA*
- 2024 **Temple-Wells Fargo Data Science Conference**, Interpretable causal inference for analyzing wearable, sensor, and distributional data *Philadelphia, PA*

Service

- 2025 **Session Chair**, Interpretable Machine Learning *INFORMS*
- 2025 **Session Chair**, Variable Importance for Scientific Discovery *INFORMS*
- 2024 **Session Chair**, Machine Learning Aided Causal Inference *INFORMS*
- 2025 **Reviewer**, Nature, European Journal of Operational Research, Neural Information Processing Systems
- 2024 **Reviewer**, Machine Learning for Health, International Conference of Machine learning
- 2023 **Reviewer**, INFORMS Journal on Computing, International Symposium on Information Theory